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1. **Wilson's disease** is a very rare disease, only about **0.005%** people are affected with it (this is an imaginary value, actual rate is even lower). Unfortunately, during a routine medical checkup, your uncle has been diagnosed with it. The best hospital in Bangladesh is **97%** effective in diagnosing this disease (if a person has it, the hospital detects it correctly). In **2%** cases the hospital **falsely detects it in a person** (the person does not actually have it). Now, you are no medical expert, but you DO know probabilistic theory. Based on the information available to you, would you suggest your **uncle starts treatment right away**, or would you suggest your uncle **performs a second test in Singapore to check if he actually has it?** Explain with mathematical reasoning. [7]

Solution:

Let **D** be the event that a person actually has Wilson's disease, and **E** be the event that the test result is positive. Then we have,

$$P(D) = 0.005\% = 5 \times 10^{-5}$$

$$P(\neg D) = 1 - P(D) = 0.99995$$

$$P(E | D) = 0.97 \text{ [Person has disease and the hospital detects positive]}$$

$$P(E | \neg D) = 0.02 \text{ [Person does not have disease but the hospital detects positive]}$$

Now, if the result is positive, probability of uncle actually having the disease can be found by:

$$\begin{aligned}
 P(D | E) &= \frac{P(E | D) P(D)}{P(E)} = \frac{P(E | D) P(D)}{P(E | D) P(D) + P(E | \neg D) P(\neg D)} = \frac{0.97 \times 5 \times 10^{-5}}{0.97 \times 5 \times 10^{-5} + 0.02 \times 0.99995} \\
 &= \frac{97}{40095} = 2.42 \times 10^{-3}
 \end{aligned}$$

Since this probability is very low, I should suggest performing a second test in Singapore.

2. Consider the following historical data on what types of cat to buy from a store. [Note: ALL cats are great, we do not discriminate. And you should adopt instead of buying.]

Colour	Age	Breed	Buy?
Orange	4	Siamese	Buy
Black	3	Siamese	Buy
White	1	Persian	Buy
Black	1	Persian	Not buy
Black	5	Siamese	Not buy
Orange	2	Siamese	Buy
Orange	3	Persian	Buy
Black	2	Siamese	Not buy
White	1	Persian	Buy
Orange	3	Siamese	Not buy

Determine whether you should buy an **orange Persian cat of age ≤ 2** using **Naive-Bayes** classifier. [8]

Solution:

$$P(\text{buy} \mid \text{orange, Persian, age} \leq 2) = P(\text{orange, Persian, age} \leq 2 \mid \text{buy}) P(\text{buy}) / P(\text{orange, Persian, age} \leq 2)$$

$$\text{Let, } 1/P(\text{orange, Persian, age} \leq 2) = a$$

$$\text{So, } P(\text{buy} \mid \text{orange, Persian, age} \leq 2) = a \times P(\text{orange, Persian, age} \leq 2 \mid \text{buy}) P(\text{buy})$$

$$\text{Now, } P(\text{buy}) = \frac{6}{10}$$

$$P(\text{orange, Persian, age} \leq 2 \mid \text{buy}) = P(\text{orange} \mid \text{buy}) P(\text{Persian} \mid \text{buy}) P(\text{age} \leq 2 \mid \text{buy}) \text{ [Naive-Bayes assumption]}$$

$$= \frac{3}{6} \times \frac{3}{6} \times \frac{3}{6} = \frac{1}{8}$$

$$\text{Hence, } P(\text{buy} \mid \text{orange, Persian, age} \leq 2) = a \times \frac{6}{10} \times \frac{1}{8} = 0.075a$$

In contrast,

$$P(\neg \text{buy} \mid \text{orange, Persian, age} \leq 2) = P(\text{orange, Persian, age} \leq 2 \mid \neg \text{buy}) P(\neg \text{buy}) / P(\text{orange, Persian, age} \leq 2)$$

$$= a \times P(\text{orange, Persian, age} \leq 2 \mid \neg \text{buy}) P(\neg \text{buy})$$

$$P(\neg \text{buy}) = \frac{4}{10}$$

$$P(\text{orange, Persian, age} \leq 2 \mid \neg \text{buy}) = P(\text{orange} \mid \neg \text{buy}) P(\text{Persian} \mid \neg \text{buy}) P(\text{age} \leq 2 \mid \neg \text{buy})$$

$$= \frac{1}{4} \times \frac{1}{4} \times \frac{2}{4} = \frac{1}{32}$$

$$\text{Hence, } P(\neg \text{buy} \mid \text{orange, Persian, age} \leq 2) = a \times \frac{4}{10} \times \frac{1}{32} = 0.0125a$$

So, I SHOULD BUY an orange Persian cat of age ≤ 2 .

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1. Your friend is facing a trial for a crime in the **District Court**. The District Court has a **true conviction rate of 98%**, meaning if a person has truly committed a crime, in 98% cases the District Court finds them guilty. In **1%** cases the District Court **falsely convicts a person** (i.e. the person has not committed the crime but is found guilty by the court). It is known that only **0.05% people in your district commit a crime on average**.

Suppose your friend has been found **guilty** in the District Court's verdict. Now, you do not know whether your friend is guilty or innocent, but you DO know probabilistic theory. Based on the information available to you, would you suggest your friend **accept the verdict**, or would you suggest they **challenge the verdict** in the High Court? Explain with mathematical reasoning. [7]

Solution:

Let **D** be the event that a person actually has committed a crime, and **E** be the event that the court finds them guilty. Then we have,

$$P(D) = 0.05\% = 5 \times 10^{-4}$$

$$P(\neg D) = 1 - P(D) = 0.9995$$

$$P(E | D) = 0.98 \text{ [Person has committed crime and the court finds guilty]}$$

$$P(E | \neg D) = 0.01 \text{ [Person has not committed crime but the court finds guilty]}$$

Now, if the the court finds my friend guilty, probability of him actually committing the crime can be found by:

$$\begin{aligned}
 P(D | E) &= \frac{P(E | D) P(D)}{P(E)} = \frac{P(E | D) P(D)}{P(E | D) P(D) + P(E | \neg D) P(\neg D)} = \frac{0.98 \times 5 \times 10^{-4}}{0.98 \times 5 \times 10^{-4} + 0.01 \times 0.9995} \\
 &= \frac{98}{2097} = 0.0467
 \end{aligned}$$

Since this probability is very low, I should suggest challenging the verdict in High Court..

2. Consider the following historical data on what types of cat to buy from a store. [Note: ALL cats are great, we do not discriminate. And you should adopt instead of buying.]

Colour	Age	Breed	Buy?
Orange	4	Siamese	Buy
Black	3	Siamese	Buy
White	1	Persian	Buy
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Black	5	Siamese	Not buy
Orange	2	Siamese	Buy
Orange	3	Persian	Buy
Black	2	Siamese	Not buy
White	1	Persian	Buy
Orange	3	Siamese	Not buy

Determine whether you should buy an **orange Persian cat of age >2** using **Naive-Bayes** classifier. [8]

Solution:

$$P(\text{buy} \mid \text{orange, Persian, age} > 2) = P(\text{orange, Persian, age} > 2 \mid \text{buy}) P(\text{buy}) / P(\text{orange, Persian, age} > 2)$$

$$\text{Let, } 1/P(\text{orange, Persian, age} > 2) = a$$

$$\text{So, } P(\text{buy} \mid \text{orange, Persian, age} > 2) = a \times P(\text{orange, Persian, age} > 2 \mid \text{buy}) P(\text{buy})$$

$$\text{Now, } P(\text{buy}) = \frac{6}{10}$$

$$P(\text{orange, Persian, age} > 2 \mid \text{buy}) = P(\text{orange} \mid \text{buy}) P(\text{Persian} \mid \text{buy}) P(\text{age} > 2 \mid \text{buy}) \text{ [Naive-Bayes assumption]}$$

$$= \frac{3}{6} \times \frac{3}{6} \times \frac{3}{6} = \frac{1}{8}$$

$$\text{Hence, } P(\text{buy} \mid \text{orange, Persian, age} > 2) = a \times \frac{6}{10} \times \frac{1}{8} = 0.075a$$

In contrast,

$$P(\neg \text{buy} \mid \text{orange, Persian, age} > 2) = P(\text{orange, Persian, age} > 2 \mid \neg \text{buy}) P(\neg \text{buy}) / P(\text{orange, Persian, age} > 2)$$

$$= a \times P(\text{orange, Persian, age} > 2 \mid \neg \text{buy}) P(\neg \text{buy})$$

$$P(\neg \text{buy}) = \frac{4}{10}$$

$$P(\text{orange, Persian, age} > 2 \mid \neg \text{buy}) = P(\text{orange} \mid \neg \text{buy}) P(\text{Persian} \mid \neg \text{buy}) P(\text{age} > 2 \mid \neg \text{buy})$$

$$= \frac{1}{4} \times \frac{1}{4} \times \frac{2}{4} = \frac{1}{32}$$

$$\text{Hence, } P(\neg \text{buy} \mid \text{orange, Persian, age} > 2) = a \times \frac{4}{10} \times \frac{1}{32} = 0.0125a$$

So, I SHOULD BUY an orange Persian cat of age > 2.